Sec: SR.IIT_*COSC(MODEL-A&APEX) | GTM-9 | Date: 18-03-26
Time: 3 Hrs | JEE-MAIN | Max. Marks: 300

KEY SHEET

PHYSICS

1	**A**	2	**A**	3	**C**	4	**C**	5	**C**
6	**C**	7	**C**	8	**D**	9	**D**	10	**C**
11	**A**	12	**D**	13	**B**	14	**B**	15	**A**
16	**D**	17	**B**	18	**A**	19	**A**	20	**A**
21	**8**	22	**3**	23	**9**	24	**5**	25	**215**

CHEMISTRY

26	**B**	27	**C**	28	**B**	29	**B**	30	**C**
31	**C**	32	**C**	33	**C**	34	**C**	35	**D**
36	**B**	37	**D**	38	**C**	39	**A**	40	**C**
41	**B**	42	**A**	43	**A**	44	**B**	45	**D**
46	**5**	47	**2.00**	48	**4**	49	**72 OR 73**	50	**6**

MATHEMATICS

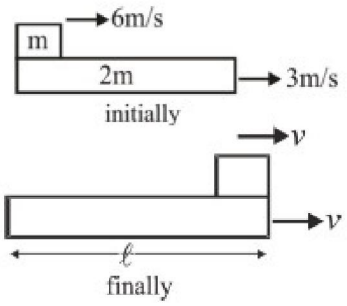
51	**C**	52	**B**	53	**C**	54	**A**	55	**B**
56	**A**	57	**A**	58	**D**	59	**A**	60	**C**
61	**C**	62	**A**	63	**D**	64	**B**	65	**C**
66	**A**	67	**B**	68	**A**	69	**C**	70	**A**
71	**13**	72	**3**	73	**3**	74	**2**	75	**3**

For More Join: @IITJEE_Advanced

SOLUTIONS

PHYSICS

1. $W_{friction} = \Delta KE$



$$-\mu mg\ell = \frac{1}{2}\,m3v^2 - \frac{1}{2}\,m6^2 - \frac{1}{2}2\,m3^2 \Rightarrow v = 4\ m/s$$

2. $$u = \frac{rd\theta}{dt}\ v = \frac{dr}{dt}\ r\frac{d\theta}{dr} = \frac{u}{v} \Rightarrow \int_{r_0}^{r_0+x}\frac{dr}{r} = \frac{v}{u}\int_0^{\pi} d\theta\ \ln\frac{r_0+x}{r_0} = \frac{v\pi}{v}\ x = r_0\left(e^{\frac{y\pi}{u}} - 1\right)\ t = \frac{x}{v} = \frac{r_0}{v}\left(e^{v\pi} - 1\right)$$

3. Explanation Question: A portion of disc is removed and we have to find centre of mass
Concept: This question is based on calculation of COM
Solution:

$$\text{mass of removed} = \frac{\sigma\pi\left(\frac{R}{2}\right)^2}{2} = \frac{\sigma\pi R^2}{8} = m$$

Centre of mass initially at $0, \frac{4R}{3\pi}$

$$\text{Now } \Delta r_{cm} = \frac{m\left(-2\left(\frac{4R/2}{3\pi}\hat{j}\right) + \frac{2R}{2}\hat{i}\right)}{4\ m} =$$

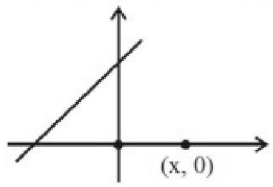
$$\frac{-R}{3\pi}\hat{j} + \frac{R}{4}\hat{i} \therefore \text{Now position} = \frac{R}{4}\hat{i} - \frac{R}{3\pi}\hat{j} + \frac{4R}{3\pi}\hat{j} = \frac{R}{4}\hat{i} + \frac{R}{\pi}\hat{j}$$

4. $$\text{Flux going in pyramid} = \frac{Q}{2\varepsilon_0}$$

Which is divided equally among all 4 faces

$$\therefore \text{ Flux through one face} = \frac{Q}{8\varepsilon_0}$$

5.



$$L_B = 2 \times L_A\ mv \times \frac{(2x+6)}{\sqrt{3^2+2^2}} = 2 \times \frac{mv \times 6}{\sqrt{3^2+2^2}}\ 2x + 6 = 12\ x = 3$$

6. $$\frac{1}{2}(M+m)\frac{m^2v^2}{(M+m)^2} = \frac{1}{2}kx^2$$

7. Question Explanation: Time for stone ($v_0 = 10\ m/s, PO = \ell = 10\ m$) thrown perpendicular to incline to hit horizontal ground.

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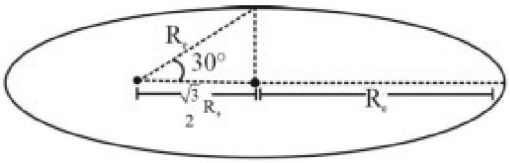
Concept: Projectile motion

Solution:

$$\Delta y = -\ell\sin 53^\circ = -8\text{ m}, v_y = v_0\sin 37^\circ = 6\text{ m/s}\ -8 = 6t - \frac{1}{2}gt^2 \text{ with } g = 10 \rightarrow 5t^2 - 6t - 8 = 0 \rightarrow t = 2\text{ s}$$

Final Answer: 2 s

8.



9. $30 + x + 60 + x = 180^\circ$

$(90 - r)\, i - r + i - r + 60 + 2r = 120$

$2i = 60^\circ \Rightarrow i = 30^\circ$

$$\frac{1}{2} = 4\sin r\ \sin r = \frac{1}{2u}\ \sin(60 - r) \geq \frac{1}{u}\frac{\sqrt{3}}{2}\cos r - \frac{1}{2}\sin ru \cdot \frac{1}{u}\frac{\sqrt{3}}{2} \times \frac{\sqrt{4u^2 - 1}}{2u} - \frac{1}{2} \times \frac{1}{2u} \geq \frac{1}{u}$$

$$\sqrt{3}\left(4u^2 - 1\right)^{1/2} \geq 5\ 3\left(4u^2 - 1\right) \geq 25\ 12u^2 - 3 \geq 25\ u^2 \geq \frac{28}{12}\ u \geq \sqrt{\frac{7}{3}}$$

10. Let r be the radius of each small droplets.

Then, volume of big drop $= 64\times$ volume of each small droplets

$$\frac{4}{3}\pi R^3 = 64 \times \frac{4}{3}\pi r^3$$

$$R = 4r\ \#(i)$$

Surface area of big drop $= 4\pi R^2$

Surface area of 64 small droplets $= 64 \times 4\pi r^2$

Increase in surface area $= 64 \times 4\pi r^2 - 4\pi R^2$

$$= 4\pi\left[64r^2 - R^2\right]$$

$$= 4\pi\left[4R^2 - R^2\right] \text{ [Using eqn.(i)]}$$

$$= 12\pi R^2$$

Energy needed = surface tension × Increase in surface area

$$= T \times 12\pi R^2 = 12\pi R^2\ T$$

11. In first case

$$\frac{\ell + 2\ell + \ldots + n\ell}{k_{eq}A} = \frac{\ell}{kA} + \frac{2\ell}{2kA} + \frac{3\ell}{3kA} + \ldots + \frac{n\ell}{nkA}$$

$$k_{eq_1} = \frac{(n+1)k}{2}$$

In second case

$$k_{eq_2} = \frac{\sum k_i A_i}{\sum A_i} = \frac{kA' + (2k)(2A') + \ldots + (nk)(nA')}{A' + 2A' + \ldots\ldots + nA}$$

$$= \frac{(2n+1)k}{3}$$

$$\frac{2n+1}{3}k = \frac{6}{5}\frac{(n+1)k}{2}$$

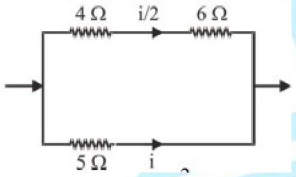
$$9n + 9 = 10n + 5$$

$$n = 4$$

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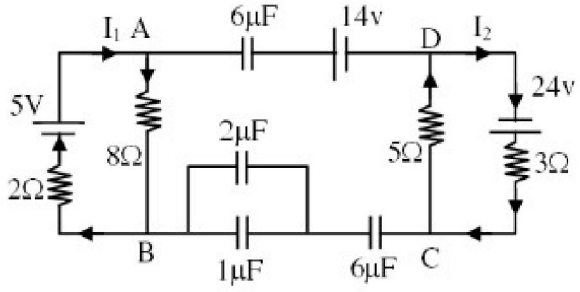
12. Explain Question:
Obtain pressure graph from given displacement graph.
Concept:
$\Delta P = -B\frac{dS}{dx}$ (pressure-displacement relation).
Solution:
Given sound wave displacement is $S(x)$.
Pressure variation is related by $\Delta P = -B\frac{dS}{dx}$
Bulk modulus B is constant, so sign and magnitude depend on slope $\frac{dS}{dx}$.
In first segment of $S(x)$ slope $\frac{dS}{dx} > 0$.
Hence $\Delta P < 0$ (negative constant).
In second segment, slope $\frac{dS}{dx} < 0$.
Hence $\Delta P > 0$ (positive constant).
In third segment, slope again $\frac{dS}{dx} < 0$.
Hence ΔP stays positive.
In last segment, slope $\frac{dS}{dx} = 0$.
Hence $\Delta P = 0$.
Thus $\Delta P(x)$ becomes a step-like graph opposite in sign to slope of $S(x)$.
This matches the rectangular graph shown in option (4).
Final answer: Option (4)

13. Since, resistance in upper branch of the circuit is twice the resistance in lower branch. Hence current there will be half.



Now $P_4 = (i/2)^2(4)\left(P = i^2R\right)$
$P_5 = (i)^2(5)$
or $\frac{P_4}{P_5} = \frac{1}{5}$
$\therefore P_4 = \frac{P_5}{5} = \frac{10}{5} = 2ca/s$

14.



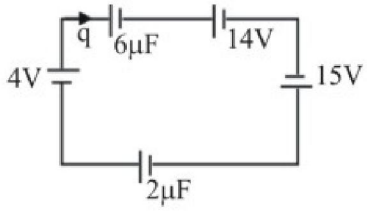
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During steady state

$I_1 = \frac{5}{10} = \frac{1}{2}$ A, $I_2 = \frac{24}{8} = 3$ A

$V_A - V_B = \frac{1}{2} \times 8 = 4v$

$V_c - V_D = 3 \times 5 = 15$ V



$q = C_{eq} V_{eq}$

$= 7.5\mu C$

$\therefore q_2 \mu F = \frac{7.5 \times 2}{2+1} = 5\mu C$

15. As the ring is in equilibrium, Net torque on the ring about the point where the ring is connected with string must be zero.

$\tau_{net} = 0$

$mgR = MB_0$ (M = magnetic dipole moment)

$mgR = I(\pi R^2)B_0$

$I = \frac{mg}{\pi R B_0}$

16. $|V| = \left|-\frac{d\phi}{dt}\right| = \oint \vec{E} \cdot \vec{dl}$

$\Rightarrow \pi b^2 \frac{B}{\Delta t} = E \times 2\pi a$

$\therefore$ Force on circular wheel

$= E \times q = E \times 2\pi a \times \lambda = \frac{\pi b^2 B}{\Delta t} \times \lambda$

Torque about centre $= a \times F = \frac{\pi b^2 B}{\Delta t} \times \lambda a = I_C \alpha$

$\therefore I_C \times \frac{\omega}{\Delta t} = \frac{\pi b^2 B \lambda a}{\Delta t}$

$\therefore \omega = \frac{\pi b^2 B \lambda a}{I_C} = \frac{\pi b^2 a B \lambda}{2I}$

17. $i^2{}_{rms} = \frac{1}{T}\int_0^T i^2 dt, i = 5t = \frac{1}{10}\int_0^{10} (5t)^2 dt$

18. $\frac{dN}{dt} \times \frac{hC}{\lambda} = P$

$F = \frac{dN}{dt}\frac{h}{\lambda} = ma$

$S = \frac{1}{2}at^2$

$S = \frac{1}{2} \times \frac{P}{mc} \times t^2$

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$$=\frac{1}{2}\times\frac{3\times10^{-3}}{50\times10^{-3}\times3\times10^{8}}\times(3600)^2$$

$$=\frac{36\times36\times10^{4}}{10^{10}}\times10^{-6}=1.3\text{ mm}$$

19. $\Delta\phi=\frac{2\pi}{\lambda}\Delta x$ & $\Delta x=\left(\frac{y}{D}\right)\times d$

$y_1=\frac{3\lambda D_1}{d}$

Now for y_2

$I_{max}\cos^2\frac{\Delta\phi}{2}=I_{max}\times\frac{3}{4}\Rightarrow\Delta\phi=\frac{\pi}{3}+2n\pi$

& this can be $4\pi+\frac{\pi}{3}$ or $6\pi-\frac{\pi}{3}6\pi-\frac{\pi}{3}$

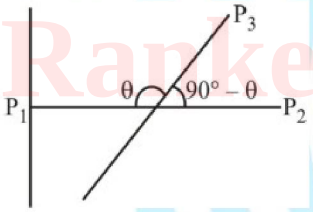
$\Rightarrow\left(\frac{y_2}{D_2}\times d\right)\frac{2\pi}{\lambda}=\Delta\phi\Rightarrow y_2=\frac{17D_2\lambda}{6\,d},\frac{13D_2\lambda}{6\,d}$

$\because y_1=y_2\Rightarrow\frac{13D_2}{6}=3D_1$

finally & $D_2=\frac{18}{13}D$

$v=\frac{D_2-D_1}{\Delta t}=\frac{5D}{13}$

20.



Let initial intensity of light is I_0. So intensity of light after transmission from first polaroid $=\frac{I_0}{2}$.

Intensity of light emitted from P_3

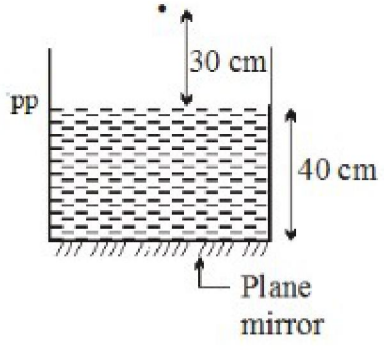
$I_1=\frac{I_0}{2}\cos^2\theta$

Intensity of light transmitted from last polaroid

$P_2=I_1\cos^2(90^\circ-\theta)$, $P_2=\frac{I_0}{2}\cos^2\theta\sin^2\theta$

$P_2=\frac{I_0}{8}(2\sin\theta\cos\theta)^2\Rightarrow P_2=\frac{I_0}{8}\sin^2 2\theta$

21.



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$v_0 = -4\ m/s \downarrow -ve$

$v_{i/m} = -\mu v_0 \Rightarrow v_{i/m} = +4/3 \times 4$

$v_{i/m} = +16/3$

$v_{i/G} = \dfrac{v_{i/m}}{\mu}$

$v_{i/G} = \dfrac{+16/3}{4/3} = 4$

$v_{i/0} = 4 - (-4) = 8\ m/s$

22. $v = \sqrt{\dfrac{\lambda x^2\ L(3g)}{L(2)\lambda x}} \Rightarrow v = \sqrt{\dfrac{3gx}{2}}$

$a = \dfrac{3g}{4}$

23. $\Delta E = \dfrac{1}{2} \times \dfrac{1 \times 0.5}{1+0.5} \times (1-(-0.5))^2 \times \left(1 - \dfrac{1}{4}\right)$

$= \dfrac{0.5}{3} \times 1.5^2 \times 1.5 \times \dfrac{3}{4} = \dfrac{4.5}{16}\ J$

$1 \times 1000 \times \Delta T = \dfrac{4.5}{16 \times 2}$

$\Delta T = \times 10^{-3}\ ^\circ C$

4. $K\varepsilon\alpha = \dfrac{Q \times m_y}{m_\alpha + m_y} \Rightarrow 48 = \dfrac{50\ m_y}{4 + m_y}$

$m_y = 96$

$\therefore$ mass of parent nuclei $= m_\alpha + m_y = 100$

25. Question Explanation: Find the thickness of a wire measured by a screw gauge with given specifications and readings.
Concept: Screw gauge reading
Solution: The pitch of the screw gauge and the number of circular scale divisions determine the least count. The thickness is the sum of the main scale reading and the circular scale reading (coinciding division x least count), assuming no zero error.
Calculation:

Least count $= \dfrac{0.25}{5 \times 100}\ cm = 5 \times 10^{-4}\ cm$

Reading $= 4 \times 0.05\ cm + 30 \times 5 \times 10^{-4}\ cm$

$= (0.2 + 0.0150)\ cm = 2.15\ mm$

Final Answer: The thickness of the wire is $2.150 \times 10^{-3}\ m$.

CHEMISTRY

26. $\alpha = \dfrac{7.8}{390} = 2 \times 10^{-2}$

$K_a = ca^2 = 16 \times 10^{-6}$ or $pK_a = 4.8$

$pK_b = 14 - 4.8 = 09.20$

27. Both NO_2^- and SCN^- are ambidentate ligands.

28. $Li > B > C > Be$ (Second ionisation enthalpy)

29. (1) $6OH^- + Br^- \rightarrow BrO_3^- + 3H_2O + 6e^-\ E_1{}^0 = -0.61, \Delta G_0{}^1 = -6 \times F(-0.61)$

(2) $2OH^- + Br^- \rightarrow BrO^- + H_2O + 2e^-\ E_2 0 = -0.76, \Delta G_0{}^1 = -2 \times F(-0.76)$

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For reaction

$BrO^- + 4OH^- \rightarrow BrO_3^- + 2H_2O + 4e^-$ $E_3^0 = ?$

$\Delta G_3^0 = -4 \times F\left(E_3^0\right)$

Target reaction, $\Delta G_3^0 = \Delta G_1^0 - \Delta G_2^0$

$-4 \times F\left(E_3^0\right) = \left[-6 \times F(-0.61)\right] - \left[-2 \times F(-0.76)\right]$

$E_3^0 = -\left[\frac{3.66 - 1.52}{4}\right] = -0.535\ V$

$\Delta G^0 = -4 \times F\left(E_3^0\right) = 206.5\ kJ.$

30. Ruby is Al_2O_3 containing about $0-1\%Cr^{+3}$ ions. In emerald, Cr^{+3} ions occupy octahedral sites in the mineral beryl.

31. $\frac{dB}{dt} = kB^2$

$\frac{1}{B_0} = \frac{1}{B_t} + kt$

$I = \frac{1}{4} + k \times 24 \Rightarrow k = \frac{1}{32}$

$\frac{1}{2} = \frac{1}{4} + \frac{1}{32} \times t \Rightarrow t = 8\ min$

32. $\left[Co\left(NH_3\right)_3 Cl_3\right]$ has no ionisable chloride ions.

33. $XeOF_2$ is T -shaped and $XeOF_4$ is square pyramidal.

34. No. of Radial nodes $= n - \ell - 1$

No. of angular nodes $= \ell$

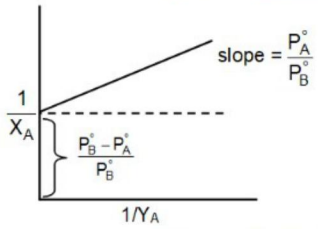
35. $\left[Cu\left(NH_3\right)_4\right]^{2+}$ hybridisation dsp^2

$Cu^{+2} - 3d^9$ has one unpaired e^-

So magnetic moment $\mu =$

$\sqrt{n(n+2} = \sqrt{1(1+2)} = \sqrt{3} = 1.73$

36.



$P_T = X_A P_A^o + X_B P_B^o = P_B^o + \left(P_A^o - P_B^o\right) X_A$

$Y_A = \frac{X_A P_A^o}{P_T}$

$\frac{P_T}{X_A} = \frac{P_A^\circ}{Y_A} \Rightarrow \frac{P_B^o + \left(P_A^o - P_B^o\right) X_A}{X_A} = \frac{P_A^o}{Y_A}$

$\frac{P_B^o}{X_A} = \left(P_B^o - P_A^o\right) + \frac{P_A^o}{Y_A}$

$\frac{1}{X_A} = \left(\frac{P_A^o}{P_B^o}\right)\frac{1}{Y_A} + \left(\frac{P_B^o - P_A^o}{P_B^o}\right)$

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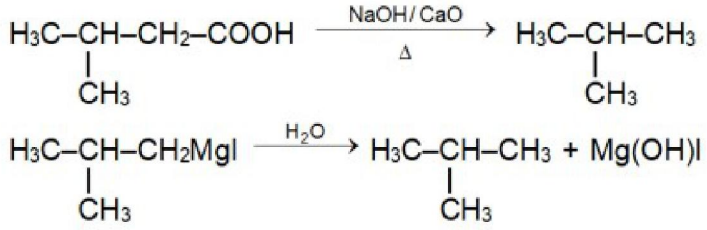
37. Rate of aromatic nucleophilic substitution reaction increases if electron withdrawing groups are attached at aromatic ring.

38. Group $\underset{-C-CH_3}{O}$ gives iodoform and 2,4-DNP test.

Esters have fruity smell and do not respond with sodium metal and $NaHCO_3$.

39. $Na, EtOH$ is used to reduce ester and aldehyde group but not amide group while $LiAlH_4$ reduces all the three groups.

40.



41. Cinnamaldehyde is obtained from cinnamon and rest are correct match.

42. pKa_H values for following are as follows

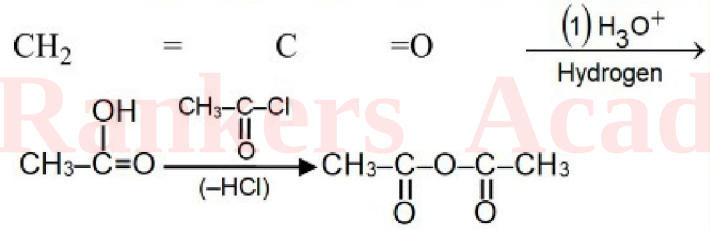
(a) $p-CH_3-C_6H_4-NH_2\ (pKa_H = 4.38)$

(b) $Et-NH_2\ (pKa_H = 10.67)$

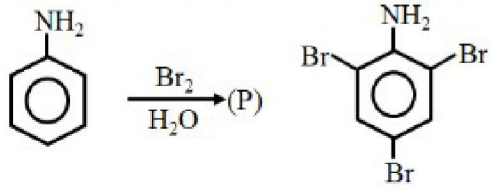
(c) $Ph-NH-CH_3\ (pKa_H = 4.84)$

(d) $Ph-N(CH_3)_2\ (pKa_H = 5.15)$

43.



44.



$-NH_2$ groups is more reactive group

2, 4, 6-Tribromo aniline (white ppt)

45. i) H_2O has highest boiling point because of H bonding.

(ii) Boiling point also depends on the magnitude of van der Waal's force of attraction, which depends on molecular weight of the compounds. Thus, the correct order is $H_2O > H_2Te > H_2Se > H_2S$.

46. $X_{glucose} = \frac{9}{109}; X_{water} = \frac{100}{109}$

$$\text{molality} = \frac{\frac{9}{109}}{\frac{100}{109} \times \frac{18}{1000}} = 5 \text{ m}$$

47. $\Delta U = 0 \Rightarrow q = -w$

$$w = -nRT\ln\frac{V_2}{V_1} = -P_1\ V_1 \ln\frac{V_2}{V_1}$$

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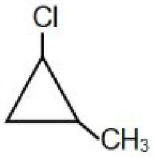
$= -10 \times 1 \ln \frac{10}{1}$

$= -10 \times 2.3$ bar-lit.

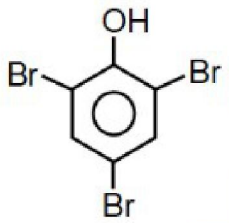
$= -23 \times 10^5 \times 10^{-3} = -2300 \text{ J} = -2.3 \text{ kJ}$

48.

$Cl-CH=CH-C_2H_5$, $Cl-CH_2-CH=CH-CH_3$,

$CH_3-C(Cl)=CH-CH_3$, 

49.

$C_6H_5NO_2 \xrightarrow[Pd]{H_2} C_6H_5NH_2 \xrightarrow[0^oC]{HNO_2} C_6H_5N_2^+ \xrightarrow[boil]{H_2O} C_6H_5OH \xrightarrow[H_2O]{Br_2 \text{ excess}}$ 

Molecular formula of $T = C_6H_3OBr_3$

Molecular mass of T=331

Mass percentage of bromine $= \frac{240}{331} \times 100 = 72.507$

50. Question Explanation: Find the magnetic moment from the total spin value.

Concept: Total Spin $S = n \times \left(\frac{1}{2}\right)$ and $\mu = \sqrt{n(n+2)}$ B.M.

Solution: The "total spin value of +3 or -3 " refers to the maximum and minimum values of the zcomponent of the total spin (M_S), which means the total spin quantum number is $S = 3$. The total spin S is related to the number of unpaired electrons (n) by the formula $S = n \times \left(\frac{1}{2}\right)$. Substituting $S = 3$, we get $3 = n \times \left(\frac{1}{2}\right)$, which gives $n = 6$ unpaired electrons. The spin-only magnetic moment (μ) is calculated using $\mu = \sqrt{n(n+2)}$. For $n = 6: \mu = \sqrt{6(6+2)} = \sqrt{6 \times 8} = \sqrt{48} \approx 6.93$ B.M.

MATHS

51. $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$

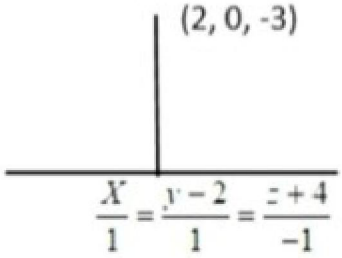
$\overrightarrow{BC} = \frac{\vec{u}}{|\vec{u}|} + \frac{\vec{v}}{|\vec{v}|} \Rightarrow \overrightarrow{AB} \cdot \overrightarrow{BC} = \left(\frac{\vec{u}}{|\vec{u}|} - \frac{\vec{v}}{|\vec{v}|}\right) \cdot \left(\frac{\vec{u}}{|\vec{u}|} + \frac{\vec{v}}{|\vec{v}|}\right)$

$= \frac{|\vec{u}|^2}{|\vec{u}|^2} - \frac{|\vec{v}|^2}{|\vec{v}|^2} = 0$

$\angle B = \frac{\pi^c}{2}$ Hence $\sum \cos 2A + 1 = 0$

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52.



53. Let $A_1, A_2 \ldots A_{2n}$ be the vertices of regular polygon of n sides .
Now select A_1 and the remaining vertives from the right hand side i.e., $n-1_{C_2}$.
Similarly on the left hand side also.
(circumcentre always lies outside the obtuse triangle)
So required probability $=\dfrac{2\times 2n\times {}^{n-1}C_2}{2n_{C_3}}$

(Here 2 n is even) or when n is odd $\dfrac{2\left(\dfrac{n-1}{2}\right)C_2.n}{2\left(n_{C_3}\right)}$

54. R be $(2\cos\theta, 3\sin\theta)$, OP perpendicular to OR

$\therefore \dfrac{3}{2}\tan\theta\times\dfrac{3}{\sqrt{3}}=-1\Rightarrow\tan\theta=\dfrac{-2\sqrt{3}}{9}$

required $\dfrac{1}{4}\left(\dfrac{1}{OP^2}+\dfrac{1}{OR^2}\right)=\dfrac{13}{144}$

55. Given $\int\tan^{-1}x\cdot\dfrac{1}{x(\ln x)}+\dfrac{1}{1+x^2}\cdot\ln(\ln x)$

$\therefore f(x)=(\tan^{-1}x)\ln(\ln x)$

56. By differentiating wrt x , we get
$(1+x^2)f(x)=x\rightarrow f(x)$ is odd

Required integral $=\int_{-\pi/4}^{\pi/4}\sec^2 x dx=2$

57. $D=p^2-5p$. If $p=9$, then D is perfect square

Area $=\int_0^9(x-9)^2dx=243$

58. n coat $\longrightarrow$ ' $35-n$ ' hooks. □ to be divided in $n-1$ places

$\dfrac{35-n}{n+1}\in N.\dfrac{36-(n+1)}{n+1}\in N\dfrac{36}{n+1}\in N$

$n+1$ possible values are $2,3,4,6,9,12,18$

59. $S=0$ as, $\sum_{r=0}^{n}n_{c_r}(-1)^r(2026)=0$

60. The given equation is $(x-1)(x-2)(x-3)=0$

61. $\cos\left(2\sin^{-1}\dfrac{1}{\sqrt{5}}\right)=\dfrac{3}{5},\sin^{-1}\left(\sin\dfrac{2\pi}{3}\right)=\dfrac{\pi}{3}$

$\cos^{-1}\left(\cos\dfrac{7\pi}{6}\right)=2\pi-\dfrac{7\pi}{6}$

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62. Use L' Hospitals rule

63. $\left|\text{adj}\left(\text{adjA}\right)\right| = |\text{A}|^{n-1^k}$

64. $f''(1) = 3 - 2a + b = af''(2)$

$= 12 - 2a = b, f'''(3) = 6 \therefore a = 3,\ b = 6$ s

65. Conceptual

66. $\int e^y dy = \int 2x dx = f(x) = \ln\left(x^2 + c\right)$

$f(1) = 1 \Rightarrow C = 1$

67. Conceptual

68. Conceptual.

69. Conceptual.

70. The number of distinct reflexive relations defined on set $A = 2^{n(n-1)}$ where $|A| = n$.

71. Conceptual

72. $(\sin\alpha\sin\beta\sin\gamma) = \dfrac{4^3(\sin\alpha\sin\beta\sin\gamma)^2}{(1+4\sin^2\alpha)(1+4\sin^2\beta)(1+4\sin^2\gamma)}$

$\Rightarrow$ either $\sin\alpha\sin\beta\sin\gamma = 0$

$\Rightarrow \sin\alpha = \sin\beta = \sin\gamma = 0$

OR

$(1+4\sin^2\alpha)(1+4\sin^2\beta)(1+4\sin^2\gamma)$

$= (4\sin\alpha)(4\sin\beta)(4\sin\gamma)$

$\Rightarrow \sin\alpha = \dfrac{1}{2} = \sin\beta = \sin\gamma$

$\cos\alpha = \pm\dfrac{\sqrt{3}}{2} = \cos\beta = \cos\gamma$

max value $\dfrac{\left(\left(\frac{1}{2}\right)+\left(\frac{1}{2}\right)+\left(\frac{1}{2}\right)\right)^2}{\left(\frac{\sqrt{3}}{2}+\frac{\sqrt{3}}{2}-\frac{\sqrt{3}}{2}\right)^2} = 3$

73. $(\sin x + 3)\left(\sin x - (x-1)^2\right) = 0$

74. Conceptual

75. $\mu = \dfrac{(40\times 12.5 - 10\times 15)}{25} = 14\sigma^2$

$= 2 + \dfrac{15\times 25}{40\times 40}(4)^2 = 5.75$

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